MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. In a parallelogram $ABCD$, $\left|\overrightarrow{AB}\right| = a$, $\left|\overrightarrow{AD}\right| = b$ and $\left|\overrightarrow{AC}\right| = c$, then $\overrightarrow{DB}.\overrightarrow{AB}$ has value

(a) $\frac{3a^2 + b^2 - c^2}{2}$ (b) $\frac{a^2 + 3b^2 - c^2}{2}$

(c) $\frac{a^2 - b^2 + 3c^2}{2}$ (d) $\frac{a^2 + 3b^2 + c^2}{2}$

2. If $A > 0$, c, d, u, v are non zero constants, and the graphs of $f(x) = |Ax + c| + d$ and $g(x) = -|Ax + u| + v$ intersect exactly at two points (1, 4) and (3, 1), then the value of $\frac{u + c}{A}$ equals

(a) 4 (b) – 4 (c) 2 (d) – 2

3. If the mapping $f(x) = ax + b$, $a > 0$ maps [–1, 1] onto [0, 2], then $\cot(\cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18)$ is

(a) $f(-1)$ (b) $f(0)$ (c) $f(1)$ (d) $f(2)$

4. If $I_1 = \int_1^{\sin\theta} \frac{x}{1+x^2}dx$ and $I_2 = \int_1^{\operatorname{cosec}\theta} \frac{dx}{x(x^2+1)}$, then the value of $\begin{vmatrix} I_1 & I_1^2 & I_2 \\ e^{I_1+I_2} & I_2^2 & -1 \\ 1 & I_1^2 + I_2^2 & -1 \end{vmatrix}$ is

(a) 1 (b) $\frac{-11}{2}$ (c) 9 (d) 0

5. If $[\sin x] + [\sqrt{2}\cos x] = -3$, $x \in [0, 2\pi]$ ($[\cdot]$ denotes the greatest integer function), then x belongs to

(a) $\left(\pi, \frac{5\pi}{4}\right)$ (b) $\left[\frac{5\pi}{4}, 2\pi\right]$ (c) $\left(\frac{5\pi}{4}, 2\pi\right)$ (d) none of these

6. The value of $\lim\limits_{x\to\infty} \frac{\cot^{-1}(x^{-a}\log_a x)}{\sec^{-1}(a^x \log_x a)} (a>1)$ is equal to

(a) 1 (b) 0 (c) $\frac{\pi}{2}$ (d) does not exist

7. Tangents PT_1 and PT_2 are drawn from a point P to the circle $x^2 + y^2 = a^2$. If the point P lies on the line $px + qy + r = 0$, then the locus of the centre of circumcircle of the triangle PT_1T_2 is

(a) $px + qy = r$ (b) $(x-p)^2 + (y-q)^2 = r^2$

(c) $px + qy = \frac{r}{2}$ (d) $2px + 2py + r = 0$

8. Let $f(x) = \lim\limits_{n\to\infty} \sum\limits_{r=0}^{n-1} \frac{x}{(rx+1)\{(r+1)x+1\}}$, then

(a) $f(x)$ is continuous but not differentiable at $x = 0$

(b) $f(x)$ is both continuous and differentiable at $x = 0$

(c) $f(x)$ is neither continuous nor differentiable at $x = 0$

(d) $f(x)$ is a periodic function

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. If $A(z_1)$, $B(z_2)$ and $C(z_3)$ are three points in argand plane where $|z_1+z_2|=|z_1|-|z_2|$ and $|(1-i)z_1+iz_3|=|z_1|+|z_3-z_1|$, then

(a) *A*, *B* and *C* lie on a fixed circle with centre $\left(\frac{z_2+z_3}{2}\right)$

(b) *A*, *B*, *C* form right angle triangle

(c) *A*, *B*, *C* form an equilateral triangle

(d) *A*, *B*, *C* form an obtuse angle triangle

10. If the sides of a triangle vary slightly in such a way that its circumradius remains constant, then $\frac{da}{\cos A}+\frac{db}{\cos B}+\frac{dc}{\cos C}$ is (where *A*, *B*, *C* are angles of a ΔABC, and *a*, *b*, *c* are sides opposite to them, *R* is circumradius)

(a) $6R$ (b) $2R$

(c) 0 (d) $2R\,(dA+dB+dC)$

11. If $f(x)=\lim\limits_{n\to\infty} e^{x\tan\left(\frac{1}{n}\right)\ln\left(\frac{1}{n}\right)}$ and $\int\frac{f(x)dx}{\sqrt[3]{\sin^{11}x\cos x}}=g(x)+C$, then

(a) $g\left(\frac{\pi}{4}\right)=\frac{3}{2}$ (b) $g(x)$ is continuous for all x

(c) $g\left(\frac{\pi}{4}\right)=\frac{-15}{8}$

(d) $g(x)$ is non differentiable at infinitely many points

12. If *A*, *B*, *C* are three points with position vectors $\hat{i}+\hat{j}, \hat{i}-\hat{j}$ and $p\hat{i}+q\hat{j}+r\hat{k}$ respectively, then the points are collinear if

(a) $p=q=r=1$ (b) $p=q=r=0$

(c) $p=q, r=0$ (d) $p=1, q=2, r=0$

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

A parabola touches co-ordinate axis at $\left(\frac{25}{3}, 0\right)$ and $\left(0, \frac{25}{4}\right)$, then

13. Focus of parabola is

(a) (4, 3) (b) (2, 1) (c) (3, 4) (d) (1, 2)

14. Length of latus rectum is

(a) $\frac{41}{5}$ (b) $\frac{44}{5}$ (c) $\frac{48}{5}$ (d) $\frac{49}{5}$

15. Equation of directrix is

(a) $4x+3y=0$ (b) $2x+y=0$ (c) $3x+4y=0$ (d) $x+2y=0$

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1.

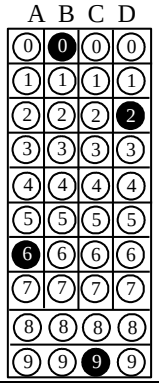
Column- I	Column -II
(A) $\lim\limits_{x\to\frac{\pi}{2}}\left[\frac{\lvert\sin x\rvert-\lvert\cos x\rvert+1}{3}\right]$ lies in (where [.] represents greatest integer function)	**1.** $(-2, 1)$
(B) f is a function satisfying $f(x+y)=f(x).f(y)$ and $f'(0)=2$. If $f\left(\frac{x}{2}\right)=kx^2$ has exactly one solution, then k is subset of	**2.** $(0, 3)$
(C) $\int_{-2}^{2}\max([x], [-x])dx$ lies in the interval (where [.] represents greatest integer function)	**3.** $(1, 3)$
(D) Let $f(x)=\cos^{-1}(\cos x)$ and $g(x)=\sin^{-1}[x+1]+\cos^{-1}[x]$ (where [.] represents greatest integer function), then the number of solution of $f(x)+g(x)=3$ lies in the interval	**4.** $(-2, 3)$ **5.** $(1, 2)$

2.

Column- I	Column -II
(A) Given that, $(1+\tan1°)(1+\tan2°)\ldots\ldots(1+\tan45°)=2^{\lambda}$, then $(\lambda+1)$ is divisible by	**1.** –2
(B) If the system of equations $x+2y+3z=7$, $2x-y+z=5$ and $x-3y-\lambda z=1$ has no solution, then λ may be	**2.** 0
(C) The value of λ for which $\frac{x-1}{2}=\frac{y+\lvert\lambda\rvert}{3}=\frac{z}{1}$ lies in the plane $2x-y-z=5$ is/ are	**3.** 3
(D) Plane $ax+y+z=a$ intersects x-axis at A, y-axis at B and z-axis at C and area of ΔABC is $\sqrt{6}$ unit, then 'a' may be	**4.** 2 **5.** –3

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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1. If $2f(x)+f(-x)=\frac{1}{x}\sin\left(\frac{x^2-1}{x}\right)$ find $\int_{1/e}^{e} f(x)\,dx$.

Space for work

mathongo

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2. The probability of a man hitting a target in one fire is $1/5$. Find the minimum number of fire; he must follow in order to make his chances of hitting the target more than $\frac{3}{4}$.

Space for work

mathongo

3. Consider two polynomials $f(x)$ and $g(x)$ as $g(x)=\sum_{r=0}^{200}\alpha_r x^r$ and $f(x)=\sum_{r=0}^{200}\beta_r x^r$.

Given (i) $\beta_r=1\,\forall r\geq 100$ (ii) $f(x+1)=g(x)$. Let $A=\sum_{r=100}^{200}\alpha_r$. Find the remainder when A is divided by 15.

Space for work

mathongo

4. The integers from 1 to 1000 are written in order around a circle. Starting at 1, every fifteenth number is marked (that is 1, 16, 31 etc.). This process is continued until a number is reached which has already been marked. If the number of marked numbers is *A*, then find $\frac{A}{50}$.

Space for work

mathongo